

ARCADENEXUS APP

Online Gaming Website

DOCUMENT PURPOSE

This Product Requirement Document (PRD) defines the core layout, brand design tokens, local persistence logic, and verification checklists for the ArcadeNexus online gaming website. It serves as the single source of truth for engineering teams and testing partners.

KEY PRODUCT SCOPE

Single-Page Dashboard: Unified header, hero space, games directory grid, scoreboard dashboard, and standardized footer.

Three Mini-Games: Tic-Tac-Toe vs Computer AI, Canvas-based Snake with speed scaling, and Card Memory Match with card flipping transitions.

Local Storage Tracking: Keeps player statistics and high scores saved on the local device browser session.

Global Audio Engine: Unified navbar switch to mute background music loop and click or game ending sound effects instantly.

1. PROJECT OVERVIEW

ArcadeNexus is a simple online gaming website. The site runs in any web browser and contains three simple games: Tic-Tac-Toe, Snake, and Memory Match. These games open and play in popup windows (modals). The website tracks user scores and total games played. It saves these scores on the user's computer using the browser's localStorage. The website is built using standard HTML, CSS, and JavaScript.

1.1 Who each deliverable is for

Deliverable File	Who it is for
index.html (Semantic Markup) / styles.css	Frontend integration developers and layout designers
app.js & games.js (JS Game Loops)	Developers verifying game controls and local state syncing
assets/ (Static Images & Audio) / README.md	Media designers updating textures; reviewers checking quickstart docs

1.2 Scope shape rationale

Pure client-side vanilla stack (HTML5, CSS3, ES6+ JS) optimized for instant offline load times. No frameworks or external server databases are permitted.

2. DELIVERABLES AT A GLANCE

#	Deliverable	Format	Exact file name
1-2	Structure & Styles	HTML5 / CSS3	`index.html` / `styles.css`
3-4	Control Logic & Games	ES6+ JavaScript	`app.js` / `games.js`
5-6	Docs & Brand Logo	Markdown / SVG	`README.md` / `assets/images/logo.svg`
7-9	Game Cover Cards	PNG Image files	`ttt_card.png` / `snake_card.png` / `memory_card.png` inside `assets/images/`
10-13	Sound Tracks & SFX	MP3 Audio files	`click.mp3` / `score.mp3` / `gameover.mp3` / `cyber_beats.mp3` inside `assets/audio/`

3. BRAND AND LOOK

Deep Void #0B0C10	Cyber Gray #1F2833	Neon Teal #66FCF1	Laser Cyan #45A29E
---	--	---	--

4. THE PRODUCT AND THE STORY

Core facts to show: (1) ArcadeNexus provides a quick-access browser games portal; (2) games operate in overlays; (3) user stats persist locally. Footer line: **Play instantly, challenge your limits on ArcadeNexus.**

4.1 Layout Beats

Beat	Section	What it does & Content Requirements
1-2	Header / Hero	Nav bar with mute control, pulsing neon slogan, CTA link.
3-5	Grid / Dashboard / Footer	3 cards with hover states, stats tables, clear button, licensing notes.

5. DELIVERABLE SPECIFICATIONS

- ``index.html``: Fully semantic layout. No inline script variables or onclick event declarations.
- ``styles.css``: CSS Variables for palette. Flexible layouts. Media breakpoints: 576px / 992px.
- ``app.js`` & ``games.js``: Binds events dynamically. Tic-Tac-Toe features active defense/attack AI. Snake features rendering loop and 5% acceleration per 5 foods. Memory Match shuffles cards and enforces 1s delays.

6. WEB QUALITY CHECKS

These checks define standard web layouts and must all pass:

1. **Direct DOM rendering:** No external JS frameworks (jQuery, React, Vue) can be imported.
2. **Semantic layout:** Uses correct standard structure (header, main, section, footer).
3. **Strict separation:** Event listeners must be bound dynamically in JS script files, not in HTML attributes.
4. **Zero errors:** Opening index.html directly from directory triggers no runtime or resource errors in Chrome.
5. **Responsive fluidity:** Under 576px wide, catalog shifts to single columns without horizontal viewport scrolling.

7. AUDIO SPECIFICATIONS

- **cyber_beats.mp3 (BGM):** Ambient loop sound track set at low volume. Plays loops automatically when toggled active.
- **UI SFX:** Click SFX (`click.mp3`), score updates (`score.mp3`), game over triggers (`gameover.mp3`) mixed at standard volume.
- **Sound Settings Routing:** Audio elements must route through mute controls globally via the navbar mute toggle.
- **Licensing:** All assets must be commercial-friendly/Creative Commons, documented in README.md.

8. FILE NAMES AND FOLDER LAYOUT

```
ArcadeNexus/  
  index.html  
  styles.css  
  app.js  
  games.js  
  README.md  
  assets/  
    images/ [logo.svg, ttt_card.png, snake_card.png, memory_card.png]  
    audio/  [click.mp3, score.mp3, gameover.mp3, cyber_beats.mp3]
```

9. WHAT MUST BE TRUE TO ACCEPT

The delivery is accepted only when all of these are true:

- Unzipped folder is named exactly `ArcadeNexus` and has all files present.
- No off-palette colors show up on visual card surfaces, headers, or text.
- Game modes (TTT, Snake, Memory) load and play smoothly inside overlay modals.
- Stats dashboard persists values using localStorage under the key name `arcadenexus\_stats`.
- Muting sound toggle globally disables background music and click/score/gameover SFX.
- Every rubric item in rubrics.txt passes with verified local execution.